

UNIT I: Foundational Models and Physical Layer

2-Mark

1. What is ARPANET?
2. What is the difference between a Static Web Page and Dynamic Web Page?

6-Mark

1. Explain the following guided transmission media: i) Twister Pair, ii) Coaxial Cable, and iii) Fiber Optics.
2. Explain the functions of hubs, switches, bridges, routers, and gateways.
3. Compare Connection-oriented and connectionless service in detail.
4. Compare and contrast the OSI model and TCP/IP model
5. Explain the layered tasks of the Network model.
6. Compare and contrast different data topologies.

Unit II: DATA LINK LAYER

2-Mark

1. What is MAC addressing?
2. Define Error Detection and Correction or explain the purpose of parity bits.
3. What is channelization in Networking?

6-Mark

1. Describe the operation of CSMA/CA in wireless LAN.
2. Explain the operation of IEEE 802.3 protocol (Standard Ethernet).
3. Describe the operation of slotted ALOHA protocol.
4. Explain the types of Physical address in Data Link Layer.
5. Describe the concept of channelization in data communication.

UNIT III: Network Layer

2-Mark

1. Differentiate subnetting with subnet mask in Networking.
2. What are the responsibilities of the Network Layer?
3. What is traffic shaping?

6-Mark

1. Explain in detail about subnetting and super netting in Computer Networks.
2. What is the need of classful IP addressing? Discuss in detail about classes of IP addresses used in Network layer.
3. Differentiate IPv4 and IPv6 along with address formats.
 4. Describe dual-stack and tunnelling techniques for IPv6 transition.
5. Compare and contrast IPv4 and IPv6 header formats.
6. Discuss the following: i) Broadcast Routing and ii) Multicast Routing.
7. Explain the concept of traffic shaping and policing.
8. Explain the concept of fragmentation in IPv4.

UNIT IV: Transport Layer and Delay-Tolerant Networking (DTN)

2-Mark

1. Write different specifications of Bundle Protocol.
2. Describe TCP flow control mechanism.
3. What is piggybacking? How is it used in transport layer?

6-Mark

1. Explain the three-way handshake process in TCP.
2. Explain the transport services provided by the transport layer.
3. Compare and contrast connection-oriented and connectionless transport protocols.
4. Explain the role of nodes and bundles in DTN.
5. Give the format of UDP Header and Explain in Brief.
6. What is the Quality-of-Service Model? Explain.

UNIT V: Application Layer and Content Delivery

2-Mark

1. What is DNS? Mention its primary function.
2. What is anonymous FTP?

6-Mark

1. Explain the working of electronic mail. How is SMTP used in Email applications?
2. Briefly explain about IMAP and MIME (Multipurpose Internet Mail Extensions) with an example.

3. List and discuss the types of DNS records.
4. Describe the SNMP message format and protocol operations.
5. Describe the differences between client-server and P2P architectures.
6. Discuss the operation of peer-to-peer Networking.
7. Describe the difference between push and pull CDNs (Content Delivery Networks).
8. Explain HTTP.